DRÄXLMAIER

# Requirement Specifications

## Automatic production line for wire processing

PA wire pre-assembly processing

| | |
|---|---|
| Requirement Specification no.: | LH_04K_000051 |
| Version: | 1.02 |
| Date of compilation: | 18.02.2025 |

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# 2 Version information

| Version | Date | Changes | Editror |
|---|---|---|---|
| 1.00 | 24.07.2024 | • First version | Juhasz Levente,OC235 |
| 1.01 | 27.08.2024 | • Chapter 15.7 – Feeding speed has been modified to 0.1 – 20mm/sec<br>• Chapter 15.7 – Feeding quantity has been inserted<br>• Chapter 15.7 – Cooling time to be adjustable has been inserted. | Juhasz Levente,OC235 |
| 1.02 | 03.02.2025 | • Chapter 4 – Table 2 has been updated<br>• Chapter 7 – Table 4 has been updated<br>• Chapter 8 – Table 5 has been updated<br>• Chapter 10 – automatic cable cutting, electrical testing and labeling has been added<br>• Chapter 11 – Table 9 has been updated<br>• Chapter 15.1 – Scanner for material identification has been inserted<br>• Chapter 15.4 – Cutting to length has been inserted<br>• Chapter 15.10 – Electrical testing has been inserted<br>• Chapter 15.11 – Labeling has been inserted | Juhasz Levente,OC235 |

**Table 1: Version information**



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# 3 Using this document

This document describes all the requirements of the DRÄXLMAIER Group for the deliveries and services of a contractor upon submittal of a tender.



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# 4 Applicable documents

| Nr. | Name | Version | Description |
|---|---|---|---|
| 1. | General Technical Delivery Specifications (GTDS) DRÄXLMAIER Group | V2 | GTDS (ATLV) |
| 2. | Safety-Related Delivery Specification (STLV) | 01 | STLV |
| 3. | Helmolz_Remote_Support_DRAEXLMAIER | 1.1 | Contains the requirements for the remote maintenance system. |
| 4. | Spare parts list template | 20200811 | List of spare parts |
| 5. | Maintenance schedule template | 20140410 | Maintenance schedule |
| 6. | MES_OPCUA_InterfaceSpecification_CS+HV_MTO | V3.03 | Interface description |
| 8. | MFU-PFU template | V1.64 | Template for machine capability and process capability. |
| 9. | Measurement Uncertainty Device and Product Tolerance (FB_PM_Messunsicherheit-EN.xlsm) | 02.03.2021V5 | Template for documenting an MGF. |
| 10. | Versionshistorie Software | Rev01_20160412 | |
| 11. | PC63 Process IDs and Error Types | V1.46 | Definition of error types and error codes for the data protocol. |
| 12. | General Technical Delivery Specifications, annex ESD protection | V4 | |
| 13. | Assembly Fixtures Display_GUI Specifications | v1.5_09.08.16 | Specifications for the design of the GUI |
| 14. | DST604A02 | A | Equipment data sheet |

**Table 2: Applicable documents**

## Note

- The applicable documents provided by the customer are legally binding parts of RFWs, orders and assignments.
- The applicable documents provided by the customer are subject to the obligation of secrecy.
- The Contractor can inform the Client of changes or modifications in writing indicating the relevant points or can submit them for decision.
- Changes or additions may be communicated to the client in writing or submitted for decision, with reference to the relevant point.
- The regulations and guidelines of the ATLV must be adhered to!



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# 5 Key contacts DRÄXLMAIER Group

| Name | Job title | Telephone | E-mail |
|---|---|---|---|
| Mr. Juhasz Levente | Backend Technology 2 | +40 723 100146 | levente-istvan.juhasz@draexlmaier.com |
| Mr. Györi Roland | Backend Technology 2 | +40 732 723879 | roland.gyoeri@draexlmaier.com |

**Table 3: Key contacts DRÄXLMAIER Group**

# 6 General project description

A production line is to be set up for the Low voltage wiring system PA for this project. This production line will be relocated to the production site after commissioning at the Contractor's plant.

Besides the classical course of the project, intensive optimization phases, training courses for personnel and test productions have to be scheduled between the commissioning at the Contractor's plant and the relocation to the production site. In order to achieve smooth final transfer to production at the series site of the Client, training courses and test productions have to be provided again after commissioning at the production site.

# 7 Project deadlines

| Procedure | Date/period |
|---|---|
| Tender released | CW 07/2025 36/2024 |
| Offer phase | CW 08/2025 |
| Concept development phase, cost workshop, order quotation | CW 08-09/2025 |
| Award of order | CW 10/2025 |
| Availability of components | CW 10/2025 |
| Production of sample components | CW 11/2025 |
| Scheduled pre-inspection date at Contractor's plant | CW 50/2025 |
| Relocation to production site | CW 02/2026 |
| Commissioning at production site | CW 04/2026 |
| Production readiness at production site | CW 05/2026 |
| Expected final acceptance at the production site | CW 09/2026 |
| SOP DRÄXLMAIER Group (start of series production) | CW 05/2026 |
| EOP customer | 2032 |

**Table 4: Project deadlines**



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# 8 Ratios

| Ratio | Value |
|---|---|
| Total quantity during the term (including Flex) | 21.275.000 pcs. |
| Maximum quantity per year (including Flex) | 2.600.000 pcs. |
| Maximum weekly quantity (including flex) | 52.000 pcs. |
| Planned shift model | 3 Shift operation, 5 Day week; 281.000 Minutes pro Year |
| Operators per shift | 1 Person |
| Required machine cycle time */ station* | <6 sec. |
| Required overall throughput time (incl. part handling) | <42 sec.(1 end) / <84 sec.(2 ends) |
| Required system availability | >98 % |
| Setup time, specification by the principal (planned value) | ≤ 5 min |
| Max. allowed system-related malfunctions per shift | ≤ 4 |
| Max. machine-related proportion of scrap | < 1 % |
| Brief description of the upstream process chain with parts logistics | Cable cutting |
| Brief description of the downstream process chain with parts logistics | Module assembly |

**Table 5: Ratios**

# 9 Product description

## 9.1 Product characteristics

| Characteristic | Description | Notes |
|---|---|---|
| Product description | Low voltage wiring system PA | |
| Material number with change level | Ex: A174 540 46 24; A000 540 91 18 | DRX mat. / FP nr.<br>8F019906 / 846C584110;<br>8F021515 / 846C716110 |
| Product surfaces | Special surface limitations:<br>No limitation. | |
| Dimensions [mm] (Cable Length) | 300 - 1500 | |
| Total weight [g] | Cca. 80 - 500 | |
| ESD component | ☐ Yes<br>☒ No | |
| Technical cleanliness requirements | ☒ Yes<br>☐ No<br>Requirements:<br>No metallic particles >1000µm allowed on the inside neither on the outside of the connector. | In order to comply with TecSa requirements, screw heads and frame profile channels that are positioned horizontally must be covered, thus avoiding accumulation of particles. |

**Table 6: Product characteristics**



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## 9.2 Product drawing

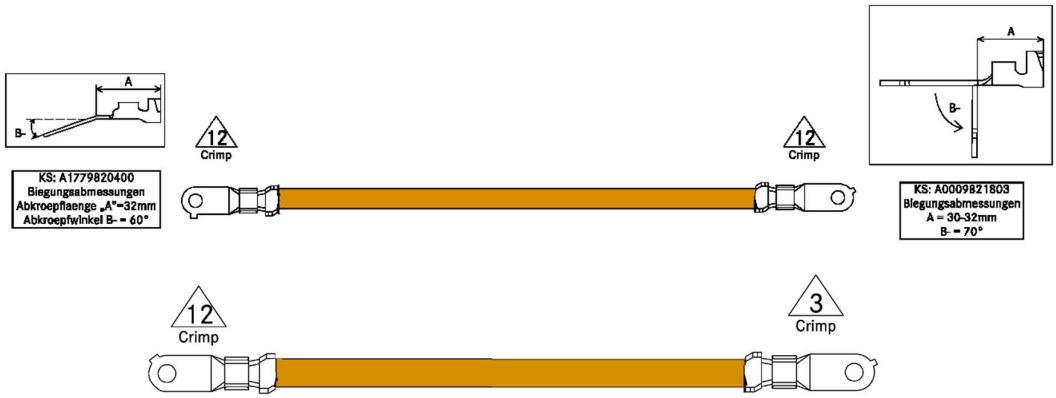



**Figure 1: Assembled product**

## 9.3 Assembly components

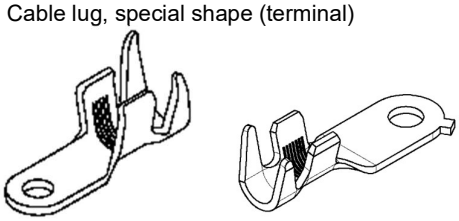
| Designation | Amount | Material number |
|---|---|---|
| Cable (Low Voltage singlecore cable 16mm²/25mm² /35mm² FLRY-B;FLR2G;)<br>  | 1 pc | L0263CXX (16mm²)<br>L3363066 (16mm²)<br>L3366063 (25mm²)<br>L3369048 (35mm²) |
| TB Cable lug, special shape (terminal)<br>  | 1 pc | A4813709<br>A4813711<br>A4813718<br>A4813719<br>A4813725<br>A4816710<br>A4816727<br>A4816743<br>A4817942<br>A4817952 |

**Table 7: Assembly components**

# 10 Process description

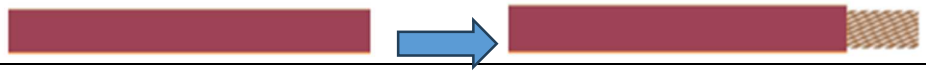
| Process description |
|---|
| ST01 - Insertion station:<br>1. Manual scanning of the identification label placed on the cable into the DRX MES.<br>• By scanning, the equipment receive the input data from DRX MES and load automatically the right processing program for the product variant<br>2. Manual insertion and positioning of the cable into the insertion unit (with possibility of buffer at least 20pcs).<br>• In case that is processed the second end of cable must ensure the right orientation between the cable lugs on the two ends.<br>• Insertion may only be possible in the correct position and the correct orientation (Poka Yoke)<br>3. Automatic fixing of cable by reaching the right position and orientation<br>4. Automatic checking of cable diameter correctness.<br>5. Automatic transfer of cable to the next station. |



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ST02 – Stripping station:

1. Automatic insert of cable end into the stripping unit.
2. Automatic stripping cut of insulation at the defined position.
3. Automatic removing of insulation part.
4. Automatic transfer of cable to the next station.



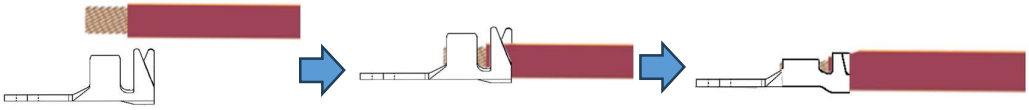
☒ Option: ST01+ST02 Cable dereeler, cutting, stripping and automatic insertion

1. Order receiving from DRX MES
2. Cable material identification by scanning
3. Automatic adjustment of cable and stripping length + stripping blade identification
4. Automatic cutting to length and stripping of cable
5. Automatic transfer of cable to the next process



ST03 – Crimping station:

1. Automatic feeding of cable lug on the crimping position
2. Automatic insert of cable end into the crimping unit
3. Automatic crimping/pressing of cable lug on cable
4. Automatic transfer of cable to the next station.



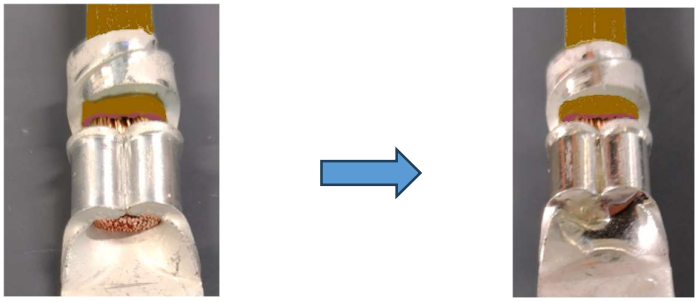
ST04 – Bending station:

1. Automatic insert of cable end with crimped cable lug into the bending unit
   - If no bending is requested by the order, the product is transferred automatically to the next station.
2. Automatic bending of cable lug on the predefined position “A” to the defined angle “B” in the defined direction “+” or “-“.
3. Automatic transfer of cable to the next station.




ST05 – Tinning station (soldering):

1. Automatic insert of cable end with crimped/bent cable lug into the tinning unit
2. Automatic tinning/soldering of the wires strands outcoming from crimped cable lug
3. Automatic cool down of product
4. Automatic transfer of cable to the next station.



ST06 – Quality inspection station:

1. Automatic insert of cable end with crimped/bent/tinned cable lug into the inspection unit
2. Automatic inspection of visual aspects of product
   - 360° vision system have to be implemented
3. Automatic transfer of cable to the next station.



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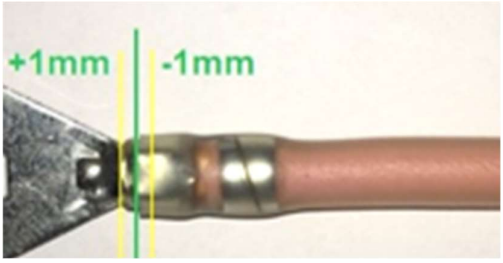
| |
|---|
| ☒ Option: ST07 – Shrinking station:<br>1. Automatic positioning of shrink tube on the cable lug<br>• If no shrinking is requested by the order, the product is transferred automatically to the next station.<br>2. Automatic insert of cable end with crimped/bent/tinned cable lug into the shrinking unit<br>3. Automatic contraction of the shrink tube<br>4. Automatic cool down of product<br>5. Automatic transfer of cable to the next station.<br>  |
| ☒ Option: ST07-1 Electrical testing<br>1. Automatic electrical checking of each product with 4wire measurement method. |
| ☒ Option: ST07-2 Labeling<br>1. Automatic attachment of an identification label to each product in the defined position |
| ST08 – Output station:<br>1. Automatic release of OK product and place on the defined position (collecting recipient)<br>2. Automatic separation of NOK product in the defined position (NOK box) |

**Table 8: Process description**

## 11 Site profile

| Characteristic | Value |
|---|---|
| Address | Romania<br>DRM Dräxlmaier Romania Sisteme Electrice S.R.L.<br>Str. Vulturului Nr. 34<br>RO-440268 Satu Mare |
| Climate | Heated hall – air-conditioned (+21 to 28°C),<br>relative air humidity up to 90% |
| Allowed construction size | Max. 11000 x 7000 x 2200mm (L x W x H)<br>(Deviations are to be specified by the supplier) |
| Allowed floor loading | Max. 8 t/m² |
| Available space | Length: 12 m; Width: 8 m; Height: 4 m |
| Max. entry portal (gates/floors) | Width: 4 m, height: 4 m |
| Width and height of aisles and transport routes | Width: 4 m, height: 4 m |
| Power supply | Max. 400V/100A or 230V/32A |
| Compressed air supply | Max. 8 bar |

**Table 9: Site profile**



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## 11.1 Hall layout

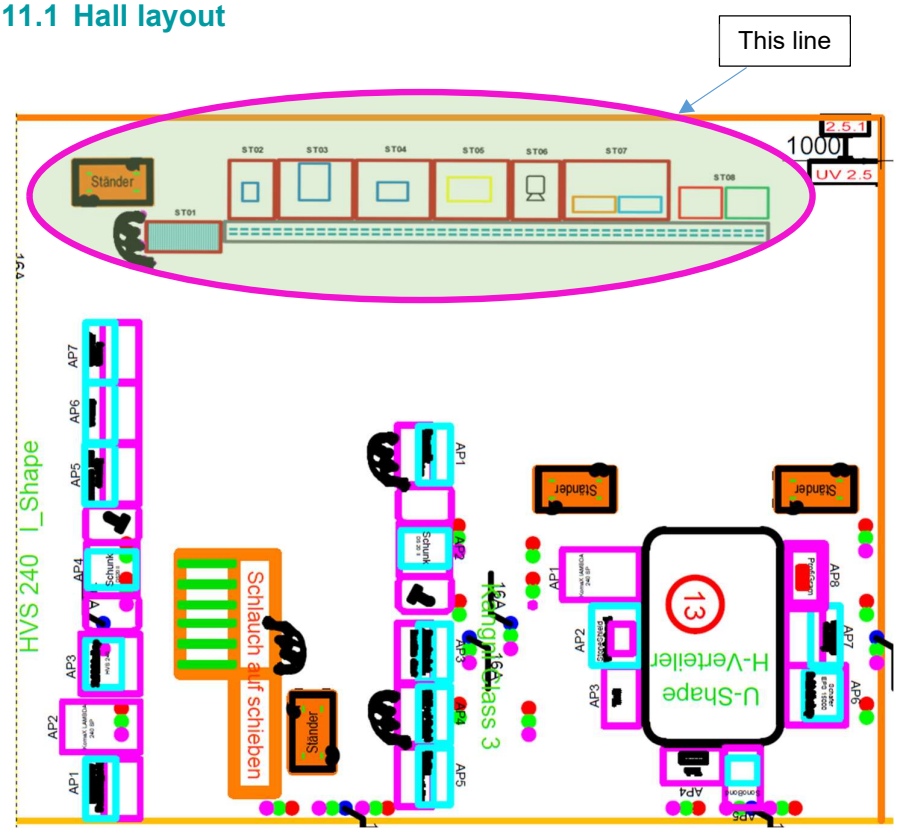



**Figure 2: Production line layout**

# 12 Layout and Design

## 12.1 General layout assumptions

1. Basic frame with integrated control, pneumatics, safety devices, etc.
   - with height-adjustable machine feet for aligning the machine bed
2. The production line has to be designed for processing all the specified components
3. Simple changing of the component inlays
   - Tool- or device changing / Poka Yoke / quick-change systems
4. Simple, ergonomic insertion/ positioning of all components from the front
5. Assure uniform linear insertion of cables in the stations.
6. Any damage to the product must be excluded.
7. All equipment and tools included in the scope of delivery must not have sharp edges and burrs, specially in the area where components are fixed and area processing
8. Any risk of injury (squeezing, cutting) during assembly must be excluded.



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9. Simple reliable technology is to be used to meet European standards as well as legal regulations from Austria and the production site.

10. No special equipment is required. The goal is the efficient, cost-effective and reliable part production in compliance with all quality criteria for the described concept.

11. ☒ Option: Starting the process by pressing a button in combination with shielding the work area with a safety light grid

12. ☒ Option: Starting the process by pressing a button in combination with shielding the work area [e.g. a hood, a protective door (sliding door), a roller blind, or similar].
    - This protective device should be closed manually and open automatically after the OK process

13. Sensors to query all components necessary for the machining process, whether they are present and correctly oriented inserted

14. All parameters / default values; such as insertion forces, holding forces, depths, under or overhang must be strictly adhered to and monitored using OPC-UA

15. The machine must automatically select the associated program based on the data's transferred from the MES with the production order. It must be possible for the client to change the data's.

## 12.2 Visualization

1. The status of the assembly station must be visualized via a ring lamp system:
    - **Green:** Assembly station is in operation or ready for production
    - **Yellow:** Warning material shortage (only for automatic feeders)
    - **Red:** Assembly station has a fault

2. The position of the ring lamp must be coordinated with the Client to the offered system concept and must not protrude from the assembly station.

3. A status lamp (red/yellow/green; see component selection) must be installed for each component mount to visualize the work cycle that has just been completed.

4. This should give the worker the information whether the component was manufactured in the OK or NOK without having to look at the GUI.
    - **Green:** Status = OK
    - **Yellow:** Status = Release process (all components inserted correctly)
    - **Red:** Status = NOK



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## 12.3 Coloring of compressed air hoses

| Use Case | Colour |
|---|---|
| Working position | Blue |
| Base position | Black |
| Main pneumatic supply | - |
| Vacuum | Yellow/ Transparent |
| Control Air | Red |
| Cooling | - |
| Heater | - |

**Table 10: Coloring of compressed air hoses**

## 12.4 Marking pneumatic parts

1. All the pneumatic parts (including hoses) have to be clearly marked.
2. The method of marking the pneumatic parts can be freely selected (e.g. numbering, labels, signs).
3. Marking only via labeling of a valve terminal is not sufficient.

## 12.5 Security of the workplace

1. All workplaces must be designed in such a way that there can be no danger (injury) of the personnel (edge protection, etc.).
2. If it is necessary from a process point of view, appropriate protective devices (e. g. noise protection, clamp protection, light protection, etc.) must be provided for the system.
3. The use of the BM in the production environment:
   - The acceptable continuous noise level is thus ≤ 70 dB(A)
   - The upper exposure limit value may not exceed 80 dB(A), or the peak value of 135 dB(C).
4. If these marginal values cannot be complied with:
   - the affected noise areas have to be recorded, marked and if possible demarcated
   - suitable measures (e.g. noise reducing programs, encapsulation) have to be described
5. Furthermore, it must be ensured that appropriate safety and hazard warnings are affixed to all relevant parts of the plant in accordance with the applicable standards and laws.
6. Sufficient contact protection shall be provided for all current and voltage-carrying parts.

### 12.5.1 Holding fixture

1. Poka Yoke Version.
2. Inclined insertion surface for easy insertion and removal of the components.
3. The holding fixture must be adapted to series parts (delay, shrinkage).



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4. The assembly station shall be designed on the tolerances of the individual parts.
5. The plastic parts are expected to be dispersed and shrinkage, which may require a sufficient process-safe fixation of the parts according to the drawing position for assembly and testing.

### 12.5.2 Sensors and mechanics

1. Any hardware required for the fixation or positioning of components must be easily accessible and adjustable and can therefore be adapted to a component distortion.
2. The hardware required for the fixing, positioning and detection of components must be painted from the Contractor after final acceptance.
3. Damage (deformation, handling traces on surfaces, ...) of the components by the assembly station must be avoided under all circumstances (sharp edges, component pads, ...).
4. An unintentional damage/adjustment of the sensors during insertion/removal of the components must be prevented by appropriate system design.

### 12.5.3 Feeding system

1. The assembly parts are supplied via plastic containers, which must be coordinated in consultation with the Client during the design approval.
2. It is necessary to create a ready parking space for the packaging of the assembly parts in the device.
3. If necessary, further feeders must be agreed with the Client.
4. Automatically feeding of the cable lug
   - Feeder capacity of at least 50pcs
   - Easy and fast possibility to exchange or setup for different cable lug types
   - Automatic identification if the actual feeding unit is the correct one for the loaded order.

## 12.6 Ergonomics

**In addition to the requirements from the GTDS (ATLV), the following issues have to be included:**

1. The range in which material is handled should not exceed the following distances:

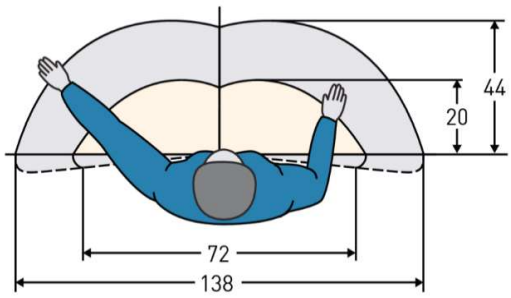



**Figure 3: Material handling range**



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2. The aim of the application of safety-related ergonomic principles for the development and design of work systems (BM in general, machines, concatenated installations, etc.) is a work structure with economic benefits that is as humane as possible.
3. By virtue of safely developed, ergonomic work conditions, the operator should be subjected to as little strain as possible and with that be able to work effectively and efficiently without fatigue!
4. Ergonomic insertion/removal (e.g. .B insertion slants) of the parts must be taken into account in the design. Particular attention must be taken to:
    - Posture while standing or sitting
    - Visual geometry (eyes - work material height)
    - Gripping geometry (gripping spaces)
    - Body forces depending on posture
    - Movement spaces and freedom of movement
5. When designing the equipment, the anthropometric requirements at the relevant serial site must be taken into account.
6. For better adaptation of the system, adjustable height must be designed. The working height must be adjustable by at least +/- 60 mm. This indication of the height adjustment range is a minimum. An extended adjustment range is desirable, whereby the actual adjustment range is to be listed in the offer. It should be noted that the working height is not the height of the table top but the actual handling area of the worker.
7. The respective working heights can be found in the current DRX GTDS (ATLV).
8. The start buttons used must be installed in an ergonomically favorable position. The selected position must be agreed with the Client.
9. Parts to be frequently adjusted, operated (e.g. tools, holders etc.) should be easily accessible, be able to be changed without tools, have positioning with repeat accuracy and be designed ergonomically.
10. The mechanical strain of the worker may not exceed 15 kg. If this is not possible, suitable aids (hoists, handling systems, etc.) are to be suggested and marked in the offer or shown separately as an option.

## 12.7 Control cabinet

1. All single conductors have to be marked.
2. The single conductors have to be able to be clearly and unmistakably allocated to the circuit diagram.
3. The single conductor labels have to be attached inside the switch cabinet.
4. Marking that is only done via labeling on a terminal strip is not sufficient.
5. The marking method of the single conductors can be freely selected (e.g. numbering, labels, signs).
6. In the control cabinet, 30% free space must remain on the hat rail (reserve).
7. For the input/output cards, 30% of the connections must remain free (reserve).
8. The design of the units has to ensure that there are no cables that run on the floor outside of the unit (e.g. between the switch cabinet and the unit), but rather that they are installed in overhead cable lines.



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9. The manufacturer is obliged to integrate air conditioning for the control cabinet if necessary.

## 12.8 Base frame

1. The base frame must be designed to be mobile incl. parking brake.
2. Attachments, maintenance units, etc. must be installed within the basic frame.
3. The maintenance unit must be accessible from the outside.
4. Interfaces to the infrastructure (plugs, couplings, ...) or suitable adapters must be provided according to the production site.
5. Interface cables must be provided with connectors at both ends of the cable in order to be able to be used flexibly.
6. The workplace must be illuminated according to the visual task, avoiding glare, flickering and large differences in brightness. The illuminance must be more than 1200 lux. LED must be selected as the light source.
7. Mobile display or several displays for visualization are to be provided, depending on the concept.
8. ☒ Option: Defined NOK storage (red scrap slide, design on component geometry, monitoring by means of sensors). Lockable and closed sideways (see component selection).
9. The system must be equipped with storage options for product folders (DIN A4, wide), reference samples per variant (component dependent) and hanging option of work instructions (DIN A4) after consultation with Client.

   In the case of an automated system, it must be closed all around (incl. cover above). On the front, a mechanical roller blind protects against contamination during standstill.
10. In the case of non-automated systems (e. g. hand lever press), a hood with storage facility must be supplied.
11. Appropriate brackets must be provided for the cables, hoses, etc. required to operate the system.
12. Powder-coated bottle holders RAL7035 (spiral form) must be used.
13. On the operator side must be a light switch for the lighting and a 230V socket implemented .
14. To be offered as an option:

    ☒ Option: Tooling and adjustment fixtures and gauges necessary for operation

    ☒ Option: A UPS for controlled shutdown of the BM control unit

    ☒ Option: 2 maintenance sockets 240V/16A, easily accessible, attached to the outside of the control cabinet, in front of the main switch of the production equipment



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## 12.9 Testing for OK - NOK

1. In case of a negative test/sensor evaluation, the processed part is declared as NOK.
2. In the case of negative visual testing (various camera systems), several test repetitions should be possible. The number of repetitions (and adjustability at the controller) must be coordinated with the AG. After reaching the upper limit, the component to be tested must be is declared as NOK.
3. Has a part been defined as NOK by the assembly station the worker may not define it as OK afterwards.
4. In case of NOK result at any station, the respective product must be not further processed on the next stations. Must be transported directly to the Output station and separated in NOK box.
5. NOK part removal may only be possible after the system has been released by authorized personnel, ex. by entering a password, key switch or similar.

## 12.10 Combined station

1. The described assembly station can produce more product variants.
2. The assembly station must be able to detect and distinguish both product variants automatically (e. g. using a sensor on the component pick-up or in the insertion fixtures).
3. Both product variants must be able to be inserted into the component pick-up or in the insertion fixtures.
4. If different component pick-ups or insertion fixtures are required, this must be coordinated with the Client.
5. If replacement equipment is required, the following requirements apply to them:
   - These must be easy to change without the help of special tools.
   - These must be able to be parked inside the assembly station. Required garages, holders or contour shots must be provided.
   - These must be labelled in such a way to ensure a unique attachment affiliation.
   - If an equipment has to be changed, the assembly station software must automatically request the operator to do so.
6. The production settings must be saved separately for each product variant.
7. The product settings for the manufactured product must be automatically loaded after selecting the product variant.

## 12.11 Process reliability

1. Default settings for monitoring tools resettable, protected by password
2. Calibration specifications: to be calibrated with the traceable measurement standard at national standard Calibration to be made by a certified testing and calibration laboratory (DKD or RENAR).
3. Inspection certificate has to be included in the delivery.
4. It has to be possible, that the calibration can be carried out by the principal (regular follow-up calibration).



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5. The following 2 user levels are to be implemented:

| Level | Role | Access protection | Activities |
|---|---|---|---|
| 0 | Machine operator | No access protection | • Operation of the system for the production of components. |
| 1 | Supervisor | Password or Key-operated switch | • Release NOK product<br>• Generate production report<br>• Check production data's on station |
| 2 | Setter | Password or Key-operated switch | • Activities out of level 0 and 1<br>• Manual mode movements<br>• Setups menus |
| 3 | Technician | Password or Key-operated switch | • All menu items in the HMI must be operable. |

**Table 11: User levels**

6. In coordination with the Client, important setting values (screw parameters, labeling data) can also be provided with password protection:
   - User: teh
   - Password: 123456

### 12.11.1 Measurement system analysis and Machine capability test

1. Process reliability must be demonstrated, documented and provided by the AN by means of machine, process and measuring equipment capability tests (MFU, PFU and MGF).
2. The test characteristics for the implementation of the capabilities must be defined in coordination between the AG and AN (DRX template shall be used).
3. Capability index Cmk ≥ 2.00 and/or Cm ≥ 2.00 (Evaluation using the attached MFU data sheet):
   - MFU pressing force +/- 20N (only at option force-displacement monitoring)
   - MFU press stroke +/- 0,1mm
   - MFU gripper X-axis +/-0,1 per transfer module
   - MFU gripper Y-axis +/-0,1 per transfer module
   - MFU Stripping length
   - MFU presses
     - MFU - pressing close size (suggested measuring device PAL 3001; Fa. C-Tec)
     - Capability index Cmk = 6,0
     - Tolerance for crimping pressing closed size: +/- 0,03mm
     - Test load 8kN
     - Data supply in .pdf und data base (MSAccess)
     - MFU crimping/pressing height
     - Tolerance for crimping/pressing height according to process specification

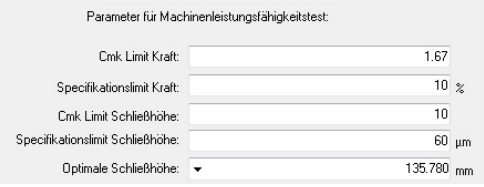



**Figure 4: Parameters for MFU**

   - Evidence PFU provided
     - Capability index Cpk ≥ 1,67 resp. Cp ≥ 1,67.



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4. Proof of measuring equipment capability "MGF" of all the integrated measuring equipment has been given (Document [Name, Version]) Capability index Cgk ≥ 1.33 and/or Cg ≥ 1.33

   **See chapter:**
   **„Applicable documents**

## 13 Component selection

1. The parts has to be in accordance with the GTDS (ATLV).
2. Components and alternatives that deviate from the following list have to be approved by the Client before they are purchased.
3. The following parts also apply for the ATLV. If there are any contradictions to the ATLV, the data given in this requirement specification shall prevail.

| **Artikel**<br>***Article*** | **Hersteller**<br>***Manufacturer*** | **Spezielle Forderungen**<br>***Special requirements*** |
|---|---|---|
| PLC control | Siemens | SIMATIC S7-1500, CPU 1511-1<br>or newer version |
| Display WAN | Siemens | KTP700 BASIC, 6AV2123-2GB03-0AX0. |
| VPN-Box | Helmholz | REX 100 WAN, 3 x LAN (Switch)/1 x |
| Initiators | Balluff | Only pluggable, M8 or M12 cable box. |
| Sensor screw depth | Meto-fer | Stop screw unit AS08/40 or similar and Sensor Element qe-osn-ps-11l – other sensors are optional selected |
| Opto sensors | Balluff, Panasonic, Omron | Only pluggable, M8 or M12 cable box. |
| Vision sensor | Keyence, Wenglor | To be agreed on with the Client. |
| System integrated scanner | Keyence | To be agreed on with the Client. |
| Camera system | Keyence, Cognex | To be agreed on with the Client. |
| Start button | Captron, Banner | |
| Robot | ABB, Kuka | To be agreed on with the Client. |
| SCARA robot | Epson | |
| Linear units | Festo, SMC | |
| Grippers (parallel grippers) | Schunk | |
| Printer (automatic applications) | Bluhm, Zebra | To be agreed on with the Client. |
| Printer (manual applications) | Zebra Technologies | ZT4xx<br>To be agreed on with the Client. |
| Screwdriver | Desoutter | |
| Aluminum profile | Item, Bosch, Robotunits,Schnaithmann | To be agreed on with the Client. |
| Vacuum pump | Becker | |



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| | | |
|---|---|---|
| Pressure and vacuum control | Balluf, Festo | To be agreed on with the Client. |
| Pressure gauge (Manometer) | Festo | MA RG (red/green area) |
| Roles | | 4x Solid swivel castors from PA/ alternatively wheel center made of PA and running surface made of polyurethane |
| Ring lamp system | WERMA | KombiSIGN70 incl. 24V LED-white/Frame BA15d |
| Status lamp | Turck | Kennleuchte K30LGRYPQ |
| Key switch to unlock the control panel | | Siemens CES SSG 10 3SB3 000–4LD01 |
| Lever cylinder to lock the NOK storage | | Lever cylinder Nr. 5634/33-MNP (Lever 92617, Gez. SSG 10) |

**Table 12: Component selection**

# 14 Functional requirements

1. Additional the assembly station hast to be equipped with a remote maintenance system. **Helmholz (REX100**) has to be used.

## 14.1 Malfunction

1. The assembly station must detect a misconduct (e. g.: breakage or wear of sensors and stamps) and visualize it as a fault.
2. Problems with the power supply (undervoltage, pressure loss,...) must lead to a shutdown/fault. It is necessary to ensure that it is not produced with undefined energy states. (Upper and lower limit value must be monitored via the SPS)
3. An error can be deleted after remediation as follows:
   - Unlocking the control panel with the key button or password
   - Pressing the acknowledge button

## 14.2 Tests

1. All testing components must test themselves cyclically, automatically. For example, by means of an additional switching process after each cycle.
2. Parameters and end positions must be monitored. The sequences of steps are also defined and are checked by the PLC.
3. If checks are missing or incorrect, the equipment goes on fault and an error message appears.
4. Testing equipment's:
   - Position / presence sensors
   - Cable diameter detection sensor
   - Crimp force analyzer
   - 360° vision system
   - Temperature measuring system
   - Crimp height measuring device
   - Pull force measuring device



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# 15 Processes

## 15.1 Scanner

1. It is not necessary to use a fixed integrated scanner. The equipment will receive the input data from DRX MES system by scanning the identification label placed on the cable into the DRX MES (scanner belong to DRX MES hardware kit).
2. The equipment must be equipped with portable scanners for material identifications.

## 15.2 Vision systems and cameras

1. If a vision system or a camera is used, they must be coordinated with the AG in advance.
2. The following topics need to be clarified in detail:
   - Inspection systematics
   - Scope
   - Resolution
   - Lighting
   - Cycle time
   - Parameterization
   - Settings

## 15.3 Transport system

1. Gripper inlays for processing without damaging, abrasion and slipping
2. Gripper pressure stepless adjustable
3. Reliable and safe transport of the contacts, insulation, cable or other components without any damaging during transfer.
4. Adjustable speed of movement
5. Exact positioning of the wire into the handling stations (adjustable)
6. Unguided wires ends must not come into the function area of the machine
7. Resolution and precision at the cross transfer positioning:
   - X-axis: 0,1mm +/-0,1mm
   - Y-axis: 0,1mm +/-0,1mm
   - Z-axis: 0,1mm +/-0,1mm
8. Monitoring of the actual shuttle position
9. In case of NOK result at any station, the respective product must be not further processed on the next stations. Must be transported directly to the Output station and separated in NOK box.

## 15.4 Cutting to length

1. Cable entry
   - Reliably inputting of the cable, cable transport mustn´t be hindered by machine parts
   - Conical straightening device, declining with at least 4 + 5 rolls, scale and quick-fastener
   - Cable feeding: min 2,5m / sec, freely programmable
   - Acceleration freely programmable



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- Length repeat accuracy:
  - Cable length smaller 500mm < +/-1mm
  - Cable length >=500mm < 0,2%
- Observing the + tolerance area has to be adjustable by correction factor
- Mechanical knot control ( diameter specific and marked adjusting disc)
- Minimum cable waste ( 1. Cable, zero cut)

2. Cut-in units
   - Pre-switching of a cable roller with cable store, types or power specified by the consultant
   - Interface description cut-in units
   - Automatic identification if the actual stripping blade is the correct one for the loaded order.

3. Length cutting
   - Length area: 500 - 2000mm
   - Cutting blades
     - Blade quality: standard materials and options, tolerances
     - Lifetime for each type

## 15.5 Stripping

1. Stripping length of each individual line 2 - 20mm, tolerance +/- 0.2mm;
   - Freely adjustable
   - Adjustable in 0,1mm steps
2. The stripping length is transferred via MES including tolerance.
3. Knifes ( specified from the supplier )
   - Stripping knifes
   - Gradation based on the wire dimensions
   - Types: V-blades, radius-V-blades, radius-blades
4. Blade quality: standard material and options
5. Lifetime per model
6. Integrated vacuum cleaning
   - No clogging of the suction hoses
   - Collection container must be designed in such a way that the container only has to be emptied once per shift (8 hours)
   - Collection container with quick change system
   - 2 collection containers should be offered

## 15.6 Pressing / Crimping

1. The AN must inform the AG of the selected crimping press
2. Admission tool head: DRX standards
   - (pressure plate with a tool feeding 45° in front to the right)
3. Manually step mode
4. Integrated crimp force analyzing
5. Quick-clamping-device with tool fixing for all standard quick-clamping tools adjustable without using any instruments
6. Designed for using shut height gages (Pal) with a standard shut height of 135.78 mm



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7. Closing force up to 22kN
8. Counter for maintenance interval (press calibration)
9. Life-time counter
10. Zero position adjustable (press calibration)
11. Press with manual crimping height adjustment
12. Mechanical press height adjustment
13. Adjustment range 1mm (+0,2/-0,8mm or +/-0,5mm), resolution 0,01mm and repeat accuracy of the closing height +/-0,01mm
14. Zero position adjustable by software
15. Adjusting range +0,2 / -0,8mm based on the zero position
16. Automatic closing height correction based on measured DCH
17. The last used closing height correction of the contact part must be automatically reloaded when the contact part is changed
18. Component-dependent material consumption recording
19. Automatic identification if the actual crimp tool is the correct one for the loaded order.

## 15.7 Bending

1. Adjustment should be done automatically by the equipment according to the order data's.
2. The bending point must be adjustable in a range of 50mm in steps of 1mm.
3. The bending angle must be adjustable in a range of 180° (from -90° till +90°) in steps of 1°
4. Is not allowed that the bending tool to damage the surface of the cable lug during the process!

## 15.8 Tinning / Soldering

1. Induction heating system should be used
2. Heating fast as possible
3. Heating temperature adjustable between 50 - 550°C with steps of 2°C
4. Heating time adjustable between 1 – 60 sec. with steps of 0.5 sec.
5. Automatic feeding of tin/soldering material (Ø0.75 – 3.00mm)
6. Feeding speed of tin/solder material adjustable between 0.1 – 20mm/sec with steps of 0.1mm/se
7. Feeding quantity of tin/solder material adjustable between 1 – 30mm with steps of 0.5mm
8. Distribution of tin/solder material on surface with adjustable movement
9. No damage of the contacts, insulation (burning, melting)
10. No tin drained over the crimp area, contact area or other part of product
11. Fast cool down of the product.
12. Cooling time adjustable between 1 – 15sec with steps of 0.5sec



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## 15.9 Shrinking / Contraction

1. Heating temperature adjustable between 100 - 300°C with steps of 1°C.
2. Contraction time adjustable between 5 – 300 sec. with steps of 1 sec.
3. Fast cool down of the product.
4. Adjustable positioning of shrink tube on the cable lug.

## 15.10 Electrical testing

1. Test type which need to perform: 4 Wire conductivity resistance measurement
2. Required result to consider as OK: R<=10 mOhm
3. Resolution 0.001mΩ
4. Preferred test equipment: Burster Resistomat 23xx
5. A calibration certificate from a ISO/IEC 17025 certified supplier shall be provided for the measurement systems.

## 15.11 Labeling

1. The equipment should be able to apply two different type of label and select the right one according to the order.
2. The type of labels:
   - Material: PP / Colour: white / Dimensions: 45mm x 100mm (DRX mat. nr. Z0300900)
   - Material: Paper / Colour: white / Dimensions: 101.6mm x 38.1mm (DRX mat. Nr. Z0300270)
3. Preferred printing type: thermal transfer with ribbon
4. The label must be applied tight around the cable on the defined position by forming a flag.
5. The margins of the label on the flag zone have to be aligned all around.
6. Position of label adjustable on a length of 30 – 150mm in steps of 1mm from the cable lug on the end with positioning precision of ±5mm.
7. Reading test of barcode/data matrix printed on label must be done automatically.

# 16 Control method / Software Requirements

## 16.1 Design

1. The software must be simple, self-explanatory and consistent to use.
2. Any future changes must simply be implemented in the SW code.
3. All cylinders must be operate/operated in a set-up mode (adjuster level).
4. Siemens TIA-Portal V.xxx (current version) is to be used for programming.



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## 16.2 HMI

### 16.2.1 General

1. The language of the visualization always has to be able to be changed between English and the national language *at the series site.*

☒ Rumanian language

### 16.2.2 Main View

1. In the main GUI view, the following points must be visualized:
   - The product name with graphical representation (3D component data)
   - Faults must be visualized as plain text in the display (efficient localization and fixing of faults/errors must be ensured)
   - Worker's instruction
     - Inserting the component(s)
     - Inserting the assembly part(s)
     - Dropping/packing/disposing
   - OK Counter of the current lot size
   - Assembly station NOK counter
   - User level/Log in
2. Display/visualization can be called up for all stations on the basic machine
3. The spelling, naming and structure of the visualization can be found in the accompanying document Vorgabe_Display_GUI_Montagevorrichtungen

⇒ **See chapter:**
**„Applicable documents**

### 16.2.3 View Production Settings

The following functions must be available in the Production Settings view (Local mode):

- Create order
- Enter the product identification data’s
- Enter the order quantity
- Enter the operator ID

### 16.2.4 Alarm archive

1. The HMI must include a "Alarm Archive" dialog.
2. In this dialog, the historical alarms must be displayed on the basis of a table.
3. The station must store the last 5000 alarms.
4. The following attributes must be stored for each alarm:
   - The date on which the alarm occurred
   - The time at which the alarm occurred
   - The time at which the alarm was acknowledged
   - The duration during which the alarm was active
   - The name of the station where the alarm occurred
   - A unique ID of the alarm
   - The severity of the alarm (Error, Warning or Info)
   - A description of the alarm



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5. The archive must provide the following filtering and sorting capabilities for each of the stored attributes:
   - Select a start and end date
   - A button for the current date ("Today")
   - Ascending sorting
   - Descending sorting
   - Filter after any text
6. An NOK archive must be set up for the last 100 components in the HMI.
7. The NOK's must be within the different assembly processes (e.g. bushing, welding, etc.) can be subdivided.
8. The archive must be able to be exported to a data carrier (e. g. USB memory stick). The file format is a Microsoft Office compatible format.

### 16.2.5 Sensor list

1. The assembly station has to provide a sensory list.
2. In this sensory list, all sensors of the assembly system must be displayed.
3. The sensors should be sorted per station.
4. The following information must be provided for each sensor:
   - Name according to the schematic
   - Description
   - Current status
5. The status of the sensors must be updated in real time.

## 16.3 Counter

### 16.3.1 Generic

1. After a restart (emergency stop, power failure, etc.) of the device, the meter readings must not be lost.
2. The counters mentioned in the following chapters must be implemented.

### 16.3.2 Counter of the packaging unit

1. In the case of a successful assembly process, the number of parts produced in this step increases after the good parts counters have been deposited.
2. In the case of assembly process with NOK result, the number of parts produced in this step increases after the NOK parts counters have been deposited.
3. After reaching the lot size, the device must lock and may only be released after a new order is created.
4. Creation of order at worker level.
5. Restarting/reload an interrupted order at Supervisor level.



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### 16.3.3 OK counter indelible

1. In the case of a successful assembly process, the number of parts produced in this step increases after the good parts counters have been deposited.
2. This counter cannot be changed or reset.
3. Visualization in all levels

### 16.3.4 NOK counter indelible

1. If a component is considered as NOK due to an NOK inspection of the system, the meter increases by the number of NOK parts.
2. This counter cannot be changed or reset.
3. Visualization in all level.

## 16.4 Interface Description, Requirements, planned Operating System

### 16.4.1 Interface Description on MES Systems of the DRÄXLMAIER Group

1. The contractor solution contains an IT system and active connection to the IT network of the principal. An interface between the production equipment (the machine) and the MES system has to be made
2. Exact requirements will be described in MES interface description that is applicable in the Requirement Specification. OPC UA should be used for the data exchange between the production equipment and the MES systems. For this, the production equipment has to provide a separate OPC UA server, to which the MES systems connect as clients.
3. Responsibility for creation and release has to be agreed to unequivocally between the principal and the contractor. This interface (connection) has to be precisely evaluated and given as an independent cost component in the offer, if necessary with options!
4. If any necessary requirements out of the MES interface description cannot be fulfilled or fulfilled only partially by the contractor, this is to be indicated in writing in the statement on the BM Requirement Specification (functional specification on the BM) and a decision is to be made regarding the next steps and documented!

### 16.4.2 Planned Operating System

1. The control and/or operating system planned for the production equipment has to be clearly defined by the contractor and agreed to by the principal.
2. If an active connection is planned to the IT network of the DRÄXLMAIER Group, it has to be coordinated with the principal.

## 16.5 Machine network and separation from DRÄXLMAIER network

### 16.5.1 Machine internal networks

1. The machine network needs to be designed, so that the control communication of components via protocols as Profinet and EtherCAT is not faulted or slowed by other protocols or programs (e.g. visualization)
2. Thereby the separation can e.g. be implemented by the usage of two machine networks (e.g. Ethernet network and Profinet network). If this is not possible, than the used switches need to have an integrated prioritization for the Profinet/EtherCAT protocols.



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### 16.5.2 Connection to MES inside the DRÄXLMAIER network

1. The machine internal networks need to be separated from the DRÄXLMAIER network. The separation is done by a firewall. Therefore a Firewall box shall be provided inside the machine, which supports port forwarding. The firewall box has to be configured with the following functions by the contractor:
   - Static IP-Address of the WAN port
     - The IP-Address, Subnet Mask & Gateway Address will be provided by the principal at the latest with commissioning
     - It shall be possible, that the principal can later carrying out changes without the contractor
2. It is not allowed that a connection is buildup from the machine network to the DRÄXLMAIER network. The connection is always buildup from the DRÄXLMAIER network into the machine network by the subsequent described port forwarding settings
3. A port forwarding has to be configured for the following applications. The used ports need to be agreed between the contractor and the principal.
   - OPC UA Server of the machine (MES connection)
   - Access to the PCs inside the machine by VNC (e.g. visualization, camera, …)
   - FTP for file exchange (e.g. pictures of a test camera)
4. Example of the network design:

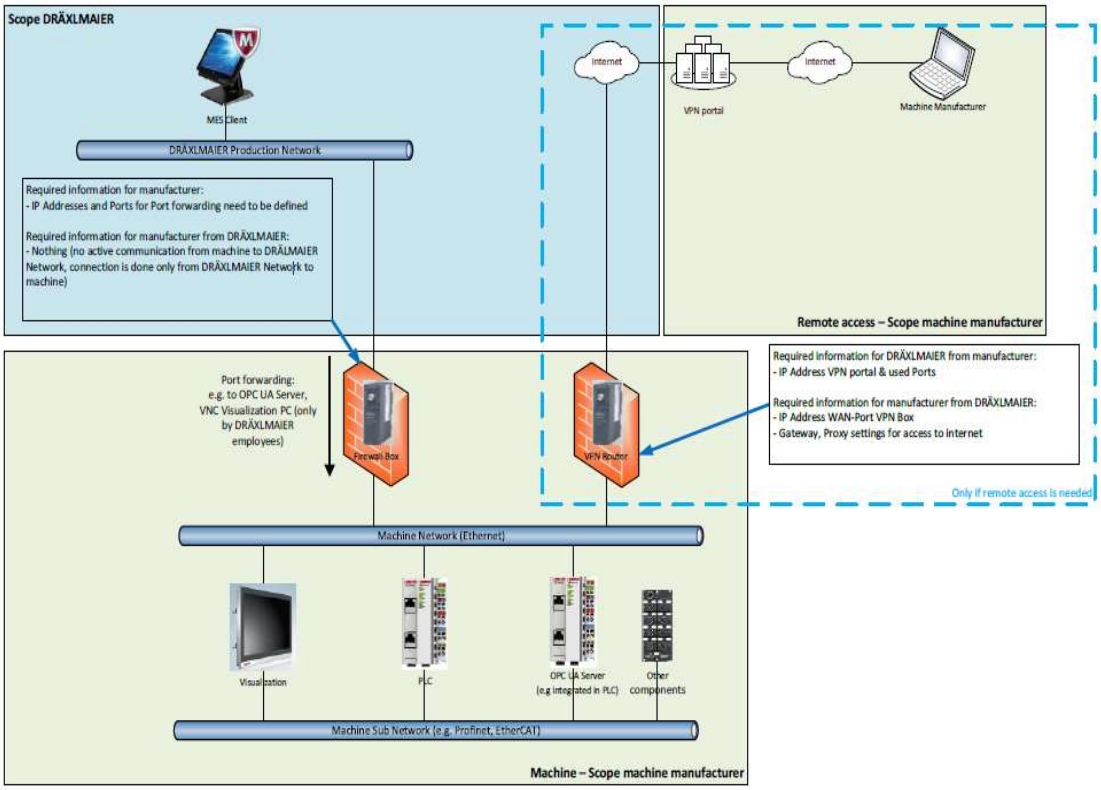



**Figure 5: Network design**



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### 16.5.3 Remote Access via VPN

1. Additionally a remote access via VPN has to be implemented into the machine. The used VPN box needs to from one of the following manufacturers:
   - Helmholz GmbH & Co. KG
   - MB connect line GmbH
2. The device must not have an integrated modem.
3. The connection of the VPN box the VPN portal should be build up by a key switch or a button in the visualization. A permanent connection is not allowed. The connection needs to be encrypted.
4. The configuration of the VPN Box needs to be done by the contractor.
5. The following information have to be handed out by the contractor to the principal before the commissioning:
   - IP-Address and Ports of the used VPN portal
6. The following information will be handed out by the principal to the contractor before the commissioning:
   - IP-Address of the WAN port
   - Subnet mask and gateway address

# 17 Purchasing conditions and offer compilation

## 17.1 Statement of cost

1. The total cost of the equipment must be broken down in the offer as follows:
   - Materials
   - Design, programming and assembly costs
   - Training, documentation and commissioning
   - Transportation costs
   - Any options

## 17.2 Terms of payment

1. Payment after final acceptance of equipment is preferred.
2. In case when divided payment method is agreed, at least 10% must be after final acceptance.

# 18 Project organization

## 18.1 Data exchange

1. The Contractor will ensure that he is capable of receiving and/or sending CAD data.

## 18.2 Documents

1. Documents are only effective if they contain the signature, date, number and version update.



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## 18.3 Contacts

1. The Contractor has to provide the following contacts for the term of the project:

| Role | Description |
|---|---|
| Project Manager | Key contact, coordinate the project |
| Lead Programmer | Is responsible for and leads development of the unit software. Contact for issues on the topic of software. |
| Designer | Is responsible for and leads the design of the unit. Contact for issues on the topic of design. |

**Table 13: Contractor's contacts**

2. The appointed people have to support the project over the whole term.
3. In principle, the respective project/implementation manager must be named as the coordinating contact person for each project (this applies to all technical and Q topics, dates, etc.)
4. All commercial points are clarified via the purchasing means of production.
5. If further contact persons (experts on the subject) are necessary, it is the responsibility of the project manager to name in writing if necessary, both internally (Client) and externally (Contractor).

## 18.4 Detailed schedule upon assignment

1. The detailed schedule (Time plan) must be drawn up by the Contractor in coordination with Client no later than 14 days after order receipt (preferably MS-Project).
2. The schedule must include at least the following:
   - Concept presentation
   - Completion Development and submission of documents for approval (construction approval)
   - Reviews / Intermediate dates in the construction phase to assess the progress of the construction:
     - Appointment 1
     - Appointment 2
     - Etc.
   - Completion Equipment, Software
   - MFU appointment at the supplier (if necessary)
   - Performance review at supplier/delivery release
   - Delivery
   - Commissioning/test phase
   - MFU date at the series site (if necessary)
   - Final acceptance
3. Component availability:
   - Reference patterns
   - Shared data
   - Components for tuning work on the plant
   - Serial components for start-up purposes
   - Components for MFU



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## 18.5 Design release

1. Design (The layout of the assembly station must be made available to the Client in advance as a DXF, STP or DWG file).
2. Process flow diagram incl. NOK processes (in coordination with Client).
3. Software process (in coordination with Client).

## 18.6 Pre-inspection/delivery release

4. Takes place at the Contractor
5. When the BM is the first of its type at the principal, than the performance review / delivery release will be made by a team from the principal at the contractor.
6. In the case of subsequent BM, this can also be done if required by an authorized specialist of the contractor (in the form of an internal contractor release process as agreed to by the principal). An appropriate written agreement between the principal and the contractor is required for this purpose.
7. Criteria for Pre-inspection / delivery release:
   - OK parts are produced
   - all checks are activated
   - all operating modes are functional (e. g..: Local operation)
   - Abuse tests (Error proving) is provided and have been passed
   - Process safety check (see point Process safety check) passed at Contractor
   - Risk assessment carried out by the contractor according to DIN EN ISO 12100, resulting measures to reduce risks have been implemented. The principal is to be granted insight into the risk assessment in the case of any concrete questions!
   - Acceptance run (criteria regarding scrap, cycle and availability must be agreed with Client)
   - Checklist Machine acceptance filled
   - Spare parts list fully filled
   - Proof of capability for equipment and all the integrated measuring equipment (MFU and MGF) filled and passed
   - Check of the scope of supply or its execution according to the specifications and ATLV. If necessary any existing deviations have to be clearly defined by the contractor.
     - The measures resulting from this are to be specified in an applicable LOP and scheduled.
   - At the latest for pre-acceptance, the media supply of the site must be simulated (frequency, voltage plug, air coupling, lighting, vibrating pots)
   - Technical documentation, technical papers on the BM are available (at least a Draft version)
   - Operating instructions are attached (at least a Draft version).
   - The conformity evaluations process (article 12 and/or 13 of the MRL) has been carried out.
   - The EG declaration of conformity has been completed and attached to the BM (Annex II of the MRL).
   - CE marking is visible, legible and permanently attached to the BM.
8. The equipment cannot be delivered to the principal until these criteria have been fulfilled!



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## 18.7 Production readiness

1. The assembly station must produce parts at the production site
2. An MFU/PFU/MGF has been passed
3. The system was approved by the safety specialist from the Client
4. However, a final acceptance is not yet possible due to defects.

## 18.8 Final acceptance

1. Takes place at the series location
2. The requirements to be fulfilled are as follows:
   - Checklist Machine acceptance filled without deviations
   - Process safety check (see point Process safety check) passed at the serial site
   - Calibration certificates of all sensors, measuring and testing equipment must be supplied
   - Calibration aids (incl. documentation) and indication of the calibration intervals must be supplied
   - Equipment documentation fully available
   - Acceptance run passed (criteria according to specifications)
   - Execution of the scope of delivery according to specifications and ATLV
   - Required measures agreed on by the contractor and the principal for the BM according to the LOP have been implemented.
   - The final acceptance report signed by an authorized representative of the contractor and the principal is available.
3. The final acceptance is considered completed if all the deviations (LOP) communicated to the contractor in writing have been solved and if all the corrected technical documents have been furnished. Any defects and functional errors not detected by the time of the inspection will not release the contractor from the duty of eliminating these within the warranty period at his own cost.

## 18.9 Training

| | | |
|---|---|---|
| The training or instruction of the production personnel takes place at the series site by: | AG Client | ☒ |
| | AN Contractor | ☐ |

**Table 14: Training location**

# 19 Documentation and marking

## 19.1 Generic

1. The documentation according to ATLV must be handed over to the specifications manufacturer in the following languages and corresponding number before final acceptance:

| | English | Romanian | Additional Languages | | |
|---|---|---|---|---|---|
| Hardcopy (DIN-A4) | 0 | 1 | | | |
| CD-ROM | 0 | 0 | | | |
| Digital | 1 | 1 | | | |

**Table 15: Documentations**



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2. The complete program / the software / control equipment of the system must be included in the system documentation in digital form.
3. The software/program must be stored on a USB stick and stored in the control cabinet in a flip box.
4. The electrical and pneumatic plans must be stored in the current version in a document bag in the control cabinet.
5. The software/program status must always be deposited with a date on the USB stick.
6. In addition, the documentation must contain the following documents:
   - CE Certificate
   - Operating Instructions
   - Maintenance plan
   - Schematics/circuit diagrams / pneumatic schemes
   - Replacement and consumable part list
   - Fixed asset BOM
   - Equipment design data in 3D (STEP)
   - Documentation of components
   - Complete drawing set in digital form
   - All programs and passwords in digital for
7. Data on lasers used:
   - Laser specifications
   - Classification
   - Type of application(s)
   - Hazard assessment/risk assessment/safety measures

## 19.2 Operating instructions

1. On the one hand, the operating instructions must include instructions on the correct (safety-relevant) behavior of the machine user for the use of the machine, taking into account foreseeable exceptional situations. On the other hand, it must contain (behavioral) information and, if necessary, further details about existing residual risks.
2. The minimum information must be as follows:
   - Information about the machine, technical data
   - Transport, installation and assembly regulations
   - Description of the machine and how it works
   - Description of the safety components (e.g. emergency stop, main switch, protective door)
   - Proper use of the machine (purpose of use)
   - Necessary or possible operator measures (e.g. marking, protective construction)
   - Notes on setting and maintenance
   - Troubleshooting and misdetection (type of error, detection, remediation)
   - Residual risks and training instructions (e.g. cutting hazard, hazardous substances)
   - Emissions of the machine (e.g. sound pressure level, gas, radiation, liquid)
   - Indication of components fed in front of main switches
   - Description of the order of commissioning processes, safety instructions, connection sketches, measures to switch off, check, monitor, correct. Notes on presetting (configuring) basic functions



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- Information on electrical power supply, electrical power consumption, characteristics for hydraulic components and pneumatic components (e.g. air consumption of the system). Information on interfaces, data transfer rates, speeds, force moments, weights and dimensions. The specified values apply to trouble-free operation. The tolerances allowed should be noted.
- A brief description of the controls and display elements. Start setting values during commissioning and possible setting values in undisturbed operation are to be described. Switching on, switching off in undisturbed operation as well as the behaviors in case of disturbed operation. The description of the error messages (fault messages, alarm messages) and description of the respective measures to be taken to eliminate them. The error messages must be meaningful and complete. The presentation is the most clearly tabular.
- List of all test equipment subject to test equipment monitoring (calibration) (as a list).
- If setup is required, a detailed setup instruction must be drawn up by the Contractor

## 19.3 Maintenance plan

1. A maintenance plan must be carried out in the xls format at the time of pre-acceptance.
2. The attached template maintenance plan document must be filled in for this purpose.

⇒ **See chapter:**
**„ Applicable documents “**

3. No reference should be made to the manufacturer's specifications.

## 19.4 List of spare parts

1. A final list of spare parts must be made at the latest at the time of pre-acceptance in the xls. format.
2. The attached Template Replacement List document must be filled in.
3. The prices of the individual components must be quoted by the AN

⇒ **See chapter:**
**„Applicable documents “**

## 19.5 Passwords and know-how protection

1. PLC programs must be secured with a password protection (know-how protection).
2. The time of handover of the passwords from the Contractor to the Client takes place after final acceptance
3. The passwords remain in the possession of the Contractor until the defined handover time
4. Until the handover date, the Client only has read access to the PLC.
5. From the time of handover, the Client has full read and write access to the PLC.
6. All passwords are handed over to the Client at the defined transfer time.
7. The following exceptions should be noted:
   - Safety Components
   - Passwords for safety components may not be transferred to the Client.
   - Standard libraries



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- Passwords for the manufacturer's standard libraries do not necessarily need to be passed.
- The affected standard libraries must be coordinated with the Client.
- If no password is passed for a standard library, a detailed interface description must be created for it and passed to the Client.

# 20 Quality requirements

## 20.1 General specifications

1. The general technical delivery specifications of the DRÄXLMAIER Group can be found in the chapter “Applicable Documents”.

⇒ **See chapter:**
**„Applicable documents“**

## 20.2 Quality monitoring

1. All components and assembly parts must be checked for the following characteristics:
   - Presence
   - Correctness of location
   - Orientation
   - Type (diameter/coding)
2. Poor assembly part quality must be detected and discharged as a scrap.
3. Monitoring of all components necessary for the assembly process for their presence, position and correct orientation

4. The 360° vision checking system must check and detect the deviations which are only visual detectable:
   - detection of bending angle
   - detection of bending position
   - detection of strands which is not covered with tin
   - detection of tin on contact surface
   - detection of strands out of crimping
   - detection of insulation crimped in crimp area
   - detection of melted insulation

5. CFA (Crimp force analyzing)
   - Realizing from 9% missing braids, as well as 50% crimped insulation (premise is a headroom of ≥ 35% and a relative variance of ≤1,0%)
   - CFA-software optimized to the provided terminals of the client
   - Default adjustments for CFA, resettable
   - The waste of the production isn´t allowed to pass 50ppm (related to the machine)
   - Press software for identification a Cpk-value (CFA). Display could be faded out. NOK-crimps doesn´t incorporate into the calculation
   - All raw data, set monitoring parameters and results must be saved and able to be transferred via an interface

6. Crimping height measuring system for crimping cable lug incl. connection to the machine software
   - Wire crimp height and tolerances to machine via MES
   - Measured values for individual parts related to MES
   - Dial into a software



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- Measuring range 0 - 20mm with a resolution of 0,001mm
- The right measure inlays are written in the process specification
- Static evaluation of the feedback
- Possibility for calibration and adjusting
- Calibration certificate with measurements, traceability (indication of the used measurement standard), indication of the measurement uncertainty
- Test laboratory has to be accredited with ISO/IEC 17025, or has to establish that it fulfills the requirements

7. Pull-off measurement system for crimping terminal
   - Hardware as component fast installed on the machine
   - Software connected on machine
   - Software-controlled
   - Measuring range 10 - 5000N (+/-1%)
   - The pull-off has to be adjusted at a defined and destroying value
   - Static evaluation of the feedback
   - Possibility for calibration and adjusting
   - Calibration certificate with measurements, traceability (indication of the used measurement standard), indication of the measurement uncertainty; Classification according to standard DIN EN ISO 7500-1
   - Test laboratory has to be accredited with ISO/IEC 17025, or has to establish that it fulfills the requirements
   - Minimum pull-out force on the pull-out measurement system via MES
   - Measured values are transferred to MES; Detailed coordination between AG and AN required

## 20.3 Environmental policy

1. The valid environmental regulations of the EU, as well as the installation site (-country) must be observed and complied with.

# 21 Service and warranty conditions

1. The AN is liable for defects for a period of 24 months without shift limitation. He is obliged to bear all necessary expenses for this purpose. This also applies in the event of a transfer of the delivery item to a foreign production site.
2. The AG shall ensure that the system(s) are handled correctly in accordance with the operating instructions and the maintenance plan. The warranty begins after successful joint final acceptance and mutual signature of a acceptance protocol.
3. If the final acceptance is negative, the warranty only begins after the necessary and successful repairs have been fulfilled within the scope of a set grace period.
4. Wear parts are excluded from the warranty/guarantee of 24 months, but are subject to the usual statutory warranties. The service life specified or agreed in a maintenance plan must be achieved.



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## 21.1 Service conditions

### 21.1.1 Requirements

In addition to the Contractors service conditions, the following requirements apply:

| To be applied yes/no | Requirements |
|---|---|
| ☒ | 1. The Contractor is obliged to maintain or set up a hotline (telephone / fax / email) in order to provide the Client with help and point out possible solutions in the event of technical problems.<br>2. The contractor must ensure that the hotline can be reached over 3 shifts on 6 working days / Monday to Friday from 08:00 to 17:00 (Time of site).<br>3. The call number or addresses and contact persons must be given to the AG immediately.<br>4. Within 1 hour of contact, the supplier must provide an answer about how to proceed. |
| ☒ | 5. The Contractor is obliged that in the event of an urgent technical problem (e.g. machine failure), a customer service representative will arrive at the series site within 24/48 hours and provide support in solving the problem. |
| ☐ | 6. A selection of wear and spare parts defined by the Client must be kept in stock in sufficient quantities. No costs may arise for the Client. |

**Table 16: Service conditions**

# 22 Parts and units provided by the customer

1. The equipment, tools, software and similar provided to the Contractor by the Client shall remain the property of the Client and shall only be made available to the Contractor free of charge for the execution of a specific order.
2. The Contractor agrees to treat them with care and to keep them in functionally flawless condition. During the period of relinquishment, the Contractor is required to adequately insure (damage, theft, fire, water,…) the relinquished objects and documents.
3. The Client will also provide a sufficient quantity of series material to the Contractor as necessary for trials and tests (see table). These parts, as stated in paragraph 2, also have to be handled carefully and used sparingly. Any materials that are not needed have to be packed according to type and returned to the Client upon completion of the order.

| Serial number | Material number | Change level | Designation | Quantity | Remark |
|---|---|---|---|---|---|
| 1. | Nodeset | | MES interface | 1 pcs | |
| 2. | MES Test Software | | | 1 pcs | |

**Table 17: Customer-provided products and software**



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# 23 Glossary

| Term | Description |
|---|---|
| Assembly station | Smaller standalone assembly and testing unit. |
| Assembly line | Assembly and test unit flexible for changes which include more stations. |
| Combined station | An assembly station that can produce 2 or more different product variants. |
| AN | Contractor |
| AG | Client / Principal |
| Product | The product that is processed by the assembly unit. |
| GTDS / ATLV | General technical delivery specifications. |
| LH/PH | Requirement specification / Specification manual. |
| Q | Quality |
| Cmk, Cpk, Cgk | Statistically calculated ratio to determine the capabilities. |
| MGF | Testing device capability (testing and measuring equipment). |
| MFU | Machine capability test |
| PFU | Process capability test |
| Display | Hardware for the display of the visualization. |
| Visualization | Diagram on the display. |
| HMI | Defines the functionality of communication between man and the unit. |
| MES | Abbreviation for the MES (Manufacturing Execution System) of the DRÄXLMAIER Group. There are different variants of the MES in the DRÄXLMAIER Group (e.g. IPST). |
| Malfunction | A malfunction is a serious error. When a malfunction occurs, the unit stops, the red warning lamp lights up and an acoustic signal is given. |
| Data protocol | A data protocol is compiled for each finished part (OK and not OK). This protocol contains e.g. the process values, test and measurement results and information on the traceability. This data protocol is send to the MES of the DRÄXLMAIER Group. The container for the data protocol (e.g. XML files, an OPC UA structure, etc.) is defined by the applied interface to the MES of the DRÄXLMAIER Group. |
| OK | The abbreviation OK defines the “OK” condition. |
| NOK | The abbreviation of not OK defines the “not OK” condition. |
| Cycle time | The cycle time is the time that elapses between the completion of the last product and the completion of the following product.<br>$\text{Cycle time} = \frac{\text{Available production time per day}}{\text{Production units required per day}}$ |
| Throughput time | The throughput time is the period of time required from the start of processing to the completion of the component. |
| Material list | The material list contains all the materials that can be installed by the assembly unit. This list is used to ensure that the orders that are delivered to the unit by the MES can also be produced. |

**Table 18: Glossary**



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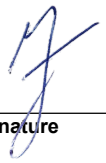
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## 26 Signature sheet

| PA wire pre-assembly processing | | | | | |
|---|---|---|---|---|---|
| Requirement specification number: | LH_04K_000051 | Requirement specification version: | 1.02 | Date of creation: | 18.02.2025 |

1. With the following, legally binding signature, the provider or the contractor recognizes the complete content of these specifications / specifications (no .: LH_04K_000051 ) and the applicable documents and undertakes to ensure that they are complied with and taken into account when submitting an offer or executing the order.
2. If the provider or the contractor has requests for changes to individual points, these must be specified in writing when submitting the offer, with reference to the corresponding point in the specifications / specifications, and must be clarified with the customer before the start of the contract.
3. Any official change that affects the order must be in writing and approved by the client.

### 26.1 Client's confirmation

| **Manufacturing Technology Backend 2 MT** | **Manufacturing Line Planning Backend SAT MLP** | **Industrial Engineering PLP** | **Industrial Engineering AE** | **Production PM** | **Quality Assurance QM** |
|---|---|---|---|---|---|
| Juhasz Levente, OC235 | Mali Szilard, OC213 | Stingaci Renata, OC-EA-SAT612 | Wagner Alexandru, OC-EA-SAT62 | Balogh Szabolcs, OC-EA-SAT1 | Muresan Ioan I., OC513 |
| **Name, Org. nr.** | **Name, Org. nr.** | **Name, Org. nr.** | **Name, Org. nr.** | **Name, Org. nr.** | **Name, Org. nr.** |
| 18.02.2025 | 18.02.2025 | 19.02.2025 | 19.02.2025 | 20.02.2025 | 20.02.2025 |
| **Date** | **Date** | **Date** | **Date** | **Date** | **Date** |
|  |  |  |  |  |  |
| **Signature** | **Signature** | **Signature** | **Signature** | **Signature** | **Signature** |

### 26.2 Contractor's confirmation

| | | |
|---|---|---|
| **Company position and department** | **Company position and department** | **Company position and department** |
| **Name with capital letters** | **Name with capital letters** | **Name with capital letters** |
| **Date** | **Date** | **Date** |
| **Signature** | **Signature** | **Signature** |



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